

POCKETSMART AI: PROJECT EXECUTABLE FILES & DEPLOYMENT RUNBOOK

1. Executive Summary & Purpose

This document catalogs all executable source code artifacts, configuration scripts, dependency specifications, and runtime commands required to deploy and execute **PocketSmart AI**. It serves as an authoritative technical runbook for system evaluators, examiners, and software engineers to reproduce, initialize, and verify the production environment from scratch.

2. Core Executable Artifacts Catalog

Executable File / Module	File Path	Language / Runtime	Primary Functionality & Responsibilities
Server Gateway & ASGI Entry Point	app.py	Python 3.12 (FastAPI)	Application initialization, CORS middleware registration, template mounting, static asset serving, and root authentication routing (/login, /register, /token, /logout).

Planning Business Logic Router	routes/planners.py	Python 3.12 (FastAPI APIRouter)	Core budgeting controllers handling requests for /home-budget, /party-budget, and /jewelry- budget, computing categorical allocations, and constructing dynamic e- commerce URLs.
Authenticatio n & Session Manager	auth.py	Python 3.12 (OAuth2 / JWT)	Password hashing (Bcrypt), JWT token synthesis, session tracking, and user query recommendatio n history logging.
AI Integration & Fallback Engine	services/gemini_utils. py	Python 3.12 (Google GenAI)	Prompt engineering interface connecting to Google Gemini generative models, complemented by the local heuristic

			fallback engine ensuring 1.2 ms zero-error responses.
Data Schema & Input Validators	<code>models/schemas.py</code>	Python 3.12 (Pydantic v2)	Strict schema definitions enforcing data integrity and runtime type validation across all incoming request payloads.
Persistent User Datastore	<code>users.json</code>	JSON Document	Atomic, flat-file persistent storage recording registered user profiles, email IDs, and cryptographic password hashes across server reloads.
Dependency Specification	<code>requirements.txt</code>	Pip Dependency Manifest	Exhaustive listing of Python libraries and locked framework versions ensuring reproducible environment provisioning.

Environment Configuration	<code>.env / .env.example</code>	Environment Variables	Secure storage for API keys (GOOGLE_API_KEY / GEMINI_API_KEY) and cryptographic session signing keys (SECRET_KEY).
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3. Step-by-Step Environment Provisioning & Execution Runbook

Step 1: System Prerequisites Verification

Ensure the target machine meets minimum runtime requirements:

- Python 3.10+ or Python 3.12 (64-bit).
- Package installer for Python (pip).
- High-speed modern web browser (Google Chrome, Microsoft Edge, or Mozilla Firefox).

Step 2: Environment Initialization & Dependency Installation

Open the project root directory in PowerShell or Terminal and execute:

1. Create an isolated virtual environment

```
python -m venv venv
```

2. Activate the virtual environment

On Windows (PowerShell):

```
.\venv\Scripts\Activate.ps1
```

On macOS / Linux:

```
source venv/bin/activate
```

3. Upgrade pip and install all locked dependencies

```
python -m pip install --upgrade pip pip install -r requirements.txt
```

Step 3: Environment Variables Configuration (.env)

Create a `.env` file in the root directory containing the required secrets:

```
GOOGLE_API_KEY=your_google_gemini_api_key_here
```

```
SECRET_KEY=pocketsmart_super_secret_jwt_key_2026
```

```
ALGORITHM=HS256
```

```
ACCESS_TOKEN_EXPIRE_MINUTES=60
```

Step 4: Launching the Production ASGI Server

Start the Uvicorn high-performance server:

```
python -m uvicorn app:app --host 0.0.0.0 --port 8000 --reload
```

4. Verification & Endpoint Access URLs

Upon server startup, the terminal confirms initialization:

INFO: Started server process [PID]

INFO: Waiting for application startup.

INFO: Application startup complete.

INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)

Interface Description	Target URL	Expected Response
Hero Landing Page	http://127.0.0.1:8000/	HTTP 200 OK (Interactive landing page rendered).
User Sign-In	http://127.0.0.1:8000/login	HTTP 200 OK (Secure login form).
User Registration	http://127.0.0.1:8000/register	HTTP 200 OK (Account creation interface).
User Dashboard	http://127.0.0.1:8000/dashboard	HTTP 200 OK (Central

		planner hub).
Home Budget Planner	http://127.0.0.1:8000/home-planner	HTTP 200 OK (Interior planner form).
Party Budget Planner	http://127.0.0.1:8000/party-planner	HTTP 200 OK (Event planning form).
Jewelry Budget Planner	http://127.0.0.1:8000/jewelry-planner	HTTP 200 OK (Multimodal vision upload form).
Interactive OpenAPI Docs	http://127.0.0.1:8000/docs	HTTP 200 OK (Swagger API documentation).

5. Production Credentials for Assessment & Defense

- **Evaluator Demo Username:** admin
- **Evaluator Demo Password:** admin123

(Alternative Master Testing Account: prabu / 1234)

6. Archival & Clean-Up Procedures

To safely stop the application and clean temporary artifacts:

1. Terminate the Uvicorn server in terminal

Ctrl + C

2. Deactivate the virtual environment

deactivate